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Towards human-AI collaboration in the competency-based curriculum development process: The case of industrial engineering and management education

Antonio Padovano *, Martina Cardamone

Department of Mechanical, Energy and Management Engineering (DIMEG), University of Calabria (UNICAL), Ponte Pietro Bucci, Building 45C, Third Floor, Arcavacata di Rende, CS, Italy

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ABSTRACT

In the endeavor to advance industrial engineering and management (IEM) education, this research underscores the imperative of supporting a dynamic and responsive adaptation of a competency-based curriculum (CBC) to meet the demands of an ever-evolving industrial landscape and job market. Our study contributes to competency-based education (CBE) by demonstrating how Artificial Intelligence (AI) can inform the definition of a CBC in the IEM field, thus initiating the pioneering steps towards a collaborative human-AI approach in CBC design. Through a stepwise methodology based on semantic analysis, text mining, natural language processing (NLP) models, informetrics approaches, and clustering algorithms, we provide data-driven insights to inform the curriculum development process. This approach enabled us to identify educational gap, particularly in domains such as digital twin engineering and human-centric IEM. Moreover, this study advocates for higher education institutions (HEIs) to embrace a more structured and collaborative approach to continuously developing competency-based curricula. In this perspective, AI (including generative AI) emerges as a valuable ally in curriculum design. This approach proves instrumental in crafting competitive and appealing curricula, especially at peripheral universities. This study culminates in an updated WING model showing how to build Industry 5.0 related curricula and a series of recommendations for engineering educators.

1. Introduction

Education plays a pivotal role in pursuing the strategic objectives outlined in the 2030 Agenda for Sustainable Development. This encompasses fostering competitive manufacturing skills, promoting social sustainability, establishing manufacturing innovation ecosystems, cultivating resilient manufacturing systems, and adapting manufacturing to the digital age. However, as computing technology, automation and artificial intelligence (AI) increasingly permeate the manufacturing industry, uncertainties arise concerning the evolving role of industrial engineers in the years ahead (World Manufacturing Forum, 2019). The Hays Global Skills Index 2019/20 (Hays, 2020) underscores the exacerbation of the talent mismatch in many labor markets, accentuating the gap between the skills possessed by jobseekers and those sought by employers. Given the remarkably dynamic and volatile nature of labor market demands, educational systems and curricula must continously adapt (Gromov et al., 2020, pp. 610–614; Gürdür Broo et al., 2022). This challenge is compounded by the absence of a clear career vision among numerous students, as many prospective job roles are yet to be conceived or developed.

Research efforts have been dedicated to evaluating the readiness of engineering education for Industry 4.0 in various countries (Ahadov et al., 2019; Jamaludin et al., 2020; Maria et al., 2018) or specific sectors (Qian et al., 2023). These studies collectively underscore the imperative to establish a cohesive roadmap for engineering education, particularly within the sphere of industrial engineering and management (IEM), to embrace the Industry 4.0 vision and, subsequently, the Industry 5.0 paradigm. Within the pursuit of aligning educational systems with ever-evolving industrial requirements, universities rigorously assess the efficacy of their curricula and persistently overhaul and expand their educational offerings.

To address these challenges, educators have turned to competency-based education (CBE) and competency-based curriculum (CBC). CBE focuses on learners' knowledge of specific topics (Hoogveld et al., 2005),

* Corresponding author.
E-mail address: antonio.padovano@unical.it (A. Padovano).

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while CBC manifests in subjects that students must master to develop the required capabilities (Sutcliffe et al., 2005). CBE offers significant advantages for higher education, especially in fields like engineering (Franco et al., 2023; Malhotra et al., 2023). Nevertheless, the curriculum development process is inherently time-consuming, often requiring input and collaboration from diverse stakeholders, including faculty members, subject matter experts, administrators, students, industry representatives, and accrediting bodies. The absence of a dynamic or structured curriculum development methodology exacerbates the complexity of this process. The resultant scalability problem gives rise to a scenario wherein educational systems either focus on highly specialized content domains manageable through routine updates, or become less sensitive to external requirements. Consequently, educators seek a scalable, dynamic, curriculum development approach, one that effectively captures pertinent knowledge domains, highlights distinctions and commonalities with other educational institutions, and provides recommendations or mappings for learning subjects.

Our study contributes to CBE by demonstrating how AI can inform the definition of a CBC in the IEM field and taking the initial steps toward a human-AI collaboration in CBC design. The methodology embodies data-driven insights generated by text mining and natural language processing (NLP) models. We have devised and implemented four integral components to empower curriculum developers when formulating and maintaining specific curricula. To validate this system, we examine the evolving landscape of the IEM discipline in response to Industry 5.0 and assess how educational systems can embrace this transformation. Subsequently, we gauge the accuracy of our proposed AI-based system through brainstorming sessions and Delphi interviews with stakeholders. This approach enables us to identify the absence of educational pathways in domains such as digital twin engineering and human-centric IEM. The remainder of the paper is structured as follows. Section 2 provides an overview of the background theory and related works. Section 3 presents the materials and methods used in this study as well as the description of the AI-based tool for curriculum development. Results of the application of the tool are presented in Section 4 and discussed in Section 5.

2. Background theory and literature review

2.1. Competency-based education theory

Competency-based education (CBE) revolves around an educational approach focused on learners' mastery of specific topics (Hoogveld et al., 2005). Sutcliffe et al. (2005) emphasize that a competency-based curriculum (CBC) materializes in a set of subjects whose objectives are to cultivate in the student a series of capacities requested by the society in which the student will operate and whose development can be demonstrated. Higher Education Institutions (HEIs) began implementing CBE in accordance with the Bologna agreements, aiming to ensure comparability standards across European university courses. This approach offers significant advantages for higher education, particularly in fields such as engineering, which demands proficiency in procedural mastery. CBE serves as a cornerstone for implementing various pedagogical methods like project-based learning and problem-based learning (Franco et al., 2023; Malhotra et al., 2023). The transition between graduates completing their studies and entering the labor market is facilitated by CBE, to the extent that employers and graduates have more information regarding what the latter are capable of doing, under the guarantee of the HEI (Daugherty et al., 2023). To achieve this, Icarte and Labate (2016) suggest a conscious alignment between competencies and curriculum content, tailoring learning units to develop competencies stated in the graduate profile.

However, literature lacks studies focused on CBC design (Henri et al., 2017). In particular, we have identified two significant gaps in the field.

– While CBE principles have been applied effectively in the medical field (Daugherty et al., 2023), their application to engineering is limited (Henri et al., 2017). Yet, nursing, medical, and engineering education share commonalities in focusing on developing technical skills, problem-solving abilities, and fostering interdisciplinary collaboration to address challenges and enhance outcomes.
– Previous CBE studies primarily relied on extensive stakeholder interactions to define curriculum requirements (Castelló et al., 2023). Although this approach ensures alignment with practical needs, it is time-consuming and lacks data-driven insights.

Our study contributes to CBE by demonstrating how AI can inform the definition of a CBC in the IEM field. The methodology integrates data-driven insights generated by text mining and NLP models. While existing research has shown the usefulness of similar methodologies in analyzing alignment between textbooks and current curricula (Yang et al., 2023), our approach leverages the latest scientific literature to design curriculum topics, ensuring alignment with the latest industry trends.

2.2. Curriculum theory and design models

Designing a competitive and appealing curriculum is not trivial, distinct from the empirical nature of physical sciences. Disagreements concerning curriculum theories often revolve around ideological considerations and are resolved through political processes (Scott, 2001). In contemporary education, two closely intertwined paradigms have emerged, each addressing the evolving needs of learners in the digital age: industry 5.0 education and education 5.0. Industry 5.0 education, as a specialized facet of industrial transformation, focuses on equipping individuals with advanced digital skills, fostering the ability to work alongside autonomous systems, and addressing the ethical and social implications of this industrial revolution. Education 5.0, on the other hand, represents a comprehensive reimagining of education itself. It transcends conventional classroom boundaries to embrace digitalization and technology integration (Grodotzki et al., 2018), adaptive and personalized learning (Bonfield et al., 2020), experiential and project-based learning (Bizami et al., 2023), and the cultivation of soft skills and multidisciplinarity (Miranda et al., 2021). Methods and tools in IEM education 5.0 are undergoing a transformative revolution. Traditional classroom instruction has evolved into a multifaceted ecosystem of personalized, learner-centric approaches. Adaptive learning platforms, driven by AI, analyze individual student progress and tailor learning experiences accordingly, providing targeted content and assessments (Ahmad et al., 2022). Additionally, virtual and augmented reality technologies immerse learners in interactive and experiential environments, transcending geographical limitations and offering hands-on experiences (Kumar et al., 2021). Experiential learning with teaching factories is also gaining popularity (Bellucci et al., 2022). When mapping these paradigms onto Hilda Taba's curriculum development model (Kaya, 2021), Industry 5.0 education primarily concerns the content and knowledge incorporated into the curriculum. In contrast, Education 5.0 focuses on how modern technologies revolutionize pedagogical methods and curriculum evaluation procedures. Curriculum development is understood as a linear process (Fig. 1), commencing with the establishment of clear objectives and progressing through content selection specified in observable or testable terms. The process concludes with the evaluation of whether those objectives have been met. Multiple stakeholders and factors significantly influence this process (Buyurgan & Kiassat, 2017). From a content perspective, curriculum theorists are concerned with the relationships between different knowledge domains within the curriculum. Two essential types of relationships are the level of integration among different knowledge domains and the progression within each domain. Progression involves determining the order in which aspects of a knowledge domain should be taught, underpinned by the belief in

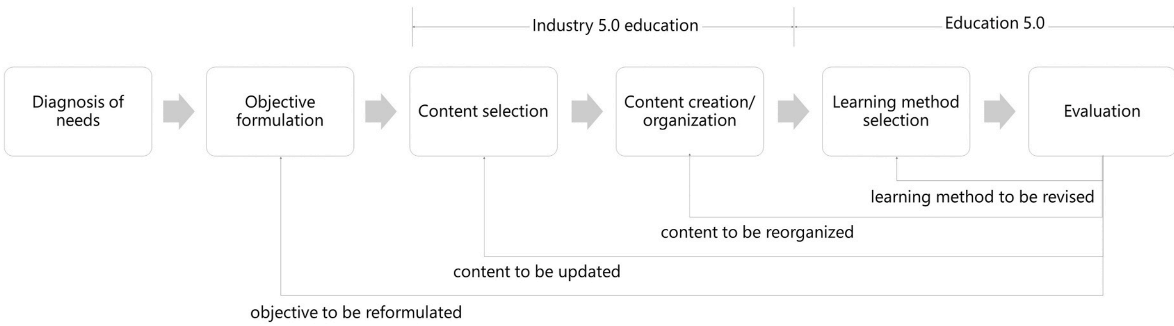

Fig. 1. Curriculum development process highlighting the difference between 'industry 5.0 education' and 'education 5.0'.

establishing a reliable hierarchy for the subject matter (Kohlberg, 1982). Curriculum theorists have developed three primary models: *curriculum as product* (Rompelman & De Graaff, 2006), *curriculum as process* (Haidet & Stein, 2006), and *curriculum as content* (Lewis, 1996). The first model entails the development of behavioral objectives, a concept frequently discussed in curriculum literature. In recent years, the idea of treating university curricula as products or engineered artifacts has gained traction (Mears et al., 2011; Rompelman & De Graaff, 2006). To do that, the four-stage process developed by Walkington (2002) represents a high-level framework, incorporating both interactive and iterative elements.

2.3. Engineering curriculum design and development

In the field of engineering education, a plethora of methodologies and tools have been employed for curriculum design and development, with an emphasis on structured and systematic approaches in many cases (Bussemaker et al., 2016; Gromov et al., 2020, pp. 610–614; Kähkönen & Hölttä-Otto, 2022; Tang, 2009, pp. 219–224). Notably, qualitative data analysis and multi-stakeholder assessments have commonly guided the definition of engineering curricula across documented works. Only a few works approached the problem in a structured way. Table 1 lists the selected studies along with their respective design methods and engineering disciplines. These studies offer valuable insights into the array of design methods and tools employed across various engineering disciplines, shedding light on their adaptability and effectiveness in addressing discipline-specific challenges and objectives. Reviewing this collection of studies and their associated design methods and disciplines, it becomes evident that there has been a discernible evolution and trend in curriculum design approaches over time. The study by Borges et al. (1994) reflects an early emphasis on expert systems as a promising tool for curriculum design, signaling the initial integration of technology into the process.

Subsequent studies by Köksal and Egìtman (1998), Boonyanuwat et al. (2008) and Mears et al. (2011) highlight the significance of stakeholder interviews and quality function deployment. This shift underscores a growing recognition of the importance of engaging various stakeholders, including students, industry professionals, and educators, in shaping the curriculum 'as-a-product'. This trend signifies a move toward more participatory and collaborative curriculum design. Recent contributions from Toral et al. (2007), Tang (2009, pp. 219–224) and Bussemaker et al. (2016) introduce concept mapping and semantic modeling, demonstrating a trend toward structured knowledge representation and organization, underpinning a more systematic and knowledge-centric approach to curriculum development. In addition to these methods, the decade 2010–2020 has witnessed the emergence of various other approaches, highlighting the growing recognition of the need for multifaceted and context-specific curriculum design strategies. Notably, the inclusion of design thinking methods is a noteworthy development, indicative of a shift toward user-centered and innovative curriculum design (Boyle et al., 2022; Fila et al., 2018; Shetty & Xu, 2018). However, it is incumbent upon future research endeavors to assess the depth and breadth of the incorporation of design thinking principles in engineering pedagogy. Furthermore, further research is essential to assess the extent to which these applications comprehensively encompass all relevant aspects of Industry 5.0. Recent years have witnessed a prominent trend in the integration of technology and data-driven approaches in curriculum design (Méndez et al., 2014). HEIs and faculty members are increasingly acknowledging the significance of data-driven decision-making in optimizing course content and tailoring curricula to specific needs. In this context, AI – mainly machine learning (ML) and text mining methods – is beginning to play a substantial role in curriculum development (Tavakoli et al., 2022). Although initial work has been conducted in specific niches, such as radiology (Weisberg & Fishman, 2020; Garg, 2020), the Internet of Things (IoT) (Somasundaram et al., 2020), or limited to exploring topics in Open Educational Resources (Molavi et al., 2020), there is substantial potential for AI-driven advancements in engineering pedagogy.

Table 1
Overview of engineering curricula development methods.

Study	Design tool or method	Discipline
Borges et al. (1994)	Expert systems	Engineering
(Köksal & Egìtman, 1998)	Stakeholder interview and quality function deployment	Industrial engineering
Toral et al. (2007)	Concept mapping	Electronic engineering
Boonyanuwat et al. (2008)	Stakeholder interview and quality function deployment	Industrial engineering
Tang (2009)	Semantic modeling, ontologies	Electrical engineering
Mears et al. (2011)	Stakeholder interview and quality function deployment	Automotive engineering
Gharaibeh et al. (2013)	Questionnaires	Telecommunications engineering
Huynh et al. (2015)	Ecological Interface Design	Engineering
Bussemaker et al. (2016)	Semantic modeling, ontologies	Chemical engineering
Wollega and Winckler (2016)	Collaborative filtering	Industrial engineering
Alarifi et al. (2016)	Body of Knowledge Analysis	Software engineering
Buyurgan and Kiassat (2017)	System engineering	Industrial engineering
Fila et al. (2018)	Design thinking	Computer, electrical and software engineering
Prasad et al. (2018)	Taba-Tyler rationale	Engineering
Shetty and Xu (2018)	Design thinking	Engineering
Kähkönen and Hölttä-Otto (2022)	Biomimetic redesign	Mechanical + Electrical Engineering
Boyle et al. (2022)	Design thinking	Manufacturing and Mechanical Engineering

To summarize, the research identifies key gaps in CBE and CBC design. A promising research avenue emerges in the realm of AI- and data-driven curriculum design. The proposed research seeks to bridge these gaps by harnessing AI to facilitate the creation, validation, and dynamic maintenance of CBC. Specifically, our focus lies within the IEM domain, providing a case study to illustrate the application of these methodologies.

3. Materials and methods

The literature analysis developed in the previous section shows that there is a notable lack of methods and practical examples on how to use AI-based and data-driven tools for curriculum design and development, which are vital for aligning educational programs with rapidly evolving industry requirements. We formulate then the following research question: 'how can AI be effectively harnessed to empower educational service providers in creating, validating, and maintaining dynamic and responsive curricula aligned with the demands of Industry 5.0?'.

3.1. AI-powered curriculum development framework

A systematic AI-powered approach to curriculum development is illustrated in Fig. 2. It extends the traditional curriculum development process (depicted in Fig. 1) and shows how AI can inform the initial stages of the process to make it more dynamic. In particular, we focus on the first three stages of the process (which are usually time-consuming) with text mining and NLP. This AI-powered curriculum development framework progresses through three distinct stages.

1. in Step 1 "Diagnosis of educational needs", the identification of relevant topics and learning items is achieved through a deep exploration of the scientific knowledge landscape and semantic analysis;
2. Step 2 "Objective formulation" is informed by an AI-powered benchmarking analysis of course syllabi at peer universities. This is necessary to ensure correct strategic positioning of the curriculum in the higher education landscape and to identify gaps or differentiation areas compared to other competing educational institutions.
3. Step 3 "Content selection" is guided by a tool recommending the priority topics and education regions (i.e. aggregation of topics) to focus on when building the curriculum based on coherence analysis.

These steps are validated through a Delphi study, engaging critical stakeholders.

1. Step 1 is validated with a group of subject matter experts in the field of Industry 4.0/5.0. These individuals possess knowledge of the sector's evolution and can confirm the skills and body of knowledge required by IEM students in the future.
2. Step 2 is validated with a group of external faculty and course directors who have insight into the positioning of universities in the educational landscape and can confirm the validity of the objectives.
3. Step 3 is validated with industry professionals and internal faculty members who can provide support in prioritizing learning content based on each institution's objectives.

The remaining three steps of the framework (content creation, learning method and evaluation) are not considered in this study, but some insights are given in Sec. 5 (e.g. discussing on the role of Generative AI for dynamic instructional content creation).

3.2. Knowledge landscape exploration and topic categorization based on semantic analysis

In the quest to determine which subjects are essential for preparing future IEM students and to comprehend the intricate tapestry of the 'industry of the future', this research relied on the maturity of topics within the scientific landscape as a barometer for their suitability and importance in curricula.

Our approach draws from informetrics (Fu et al., 2022; Mejia et al., 2021). Informetrics refers to the study of quantitative aspects of information in any form, including and beyond the academic community and academic outputs. Hence, it is an extension and evolution of traditional bibliometrics and scientometrics (Xu & Feng, 2021). To facilitate these studies, researchers utilize specialized toolkits such as the Statistical Analysis Toolkit for Informetrics or SATI (Xu & Feng, 2021) and the Content Analysis Toolkit for Academic Research[1] or CATAR (Fu et al., 2022). In particular, Fu et al. (2022) employ an approach similar to the one presented in this research. They use CATAR, a widely recognized tool in scientometrics. They employ techniques like agglomerative hierarchical clustering and multi-dimensional scaling, based on bibliographical coupling similarity, for topic identification and cross-validation. Similarly, Mejia et al. (2021) employ hierarchical topic tree (HTT) and scientific evolutionary pathways (SEP) to measure semantic similarity and track the evolution of scientific topics over time. However, as observed by the publication records retrieved from SCOPUS, we noticed a limited number of studies combining semantic analysis in the informetrics field, especially in the engineering discipline.

Scientific documents' metadata were extracted at the beginning of September 2022 from SCOPUS database using the following search string: ("industry 4.0" OR "industry 5.0") AND (manufacturing OR production). Document keywords were considered as a representation of the document content, and therefore, as elementary constituents of the knowledge landscape. 46900 unique keywords were obtained. To reduce the bias due to a wrongly typed keyword or irrelevant keywords, only keywords that appear at least twice are considered. 19744 keywords result from the frequency analysis.

To find clusters of semantically-related keywords, Sentence-BERT (Bidirectional Encoder Representations from Transformers), also known as SBERT, has been used (Reimers & Gurevych, 2019). BERT is a family of state-of-the-art language models, which has become a ubiquitous baseline in NLP studies and experiments together with Generative Pre-trained Transformers (GPT). SBERT aims to adapt the BERT architecture by using siamese and triplet network structures to derive semantically meaningful sentence embeddings. In practice, these models generate contextual embeddings of input tokens (in our case, keywords), each infused with information of its neighborhood. SBERT maps the tokens to a 384 dimensional dense vector space that can be compared later using cosine-similarity for tasks like hierarchical clustering for topic modeling or semantic textual similarity. To visualize items and topics in an interpretable way, data dimensionality was reduced using t-SNE (t-distributed Stochastic Neighbor Embedding). t-SNE is an unsupervised non-linear dimensionality reduction technique for data exploration and visualizing high-dimensional data. Unlike Principal Component Analysis (PCA), which aims to maximize variance by preserving large pairwise distances, t-SNE prioritizes retaining small pairwise similarities between data points in a lower-dimensional space. This approach provides an intuitive representation of how data is organized in higher dimensions, offering insights into underlying patterns and relationships within the dataset. In our t-SNE map, proximity between points denoted their semantic similarity. Additionally, each point's size in the map denoted the frequency of its respective keyword in the literature, thereby signifying its importance and maturity within global discourse. To further illuminate the landscape, hierarchical clustering via Ward's algorithm was executed, effectively grouping keywords into cohesive clusters. These topics, in turn, were thoughtfully organized into high-level subject areas. This comprehensive approach facilitated the identification of essential subjects for future IEM students and provided

[1] See CATAR website: https://web.ntnu.edu.tw/~samtseng/CATAR/Readme.html.

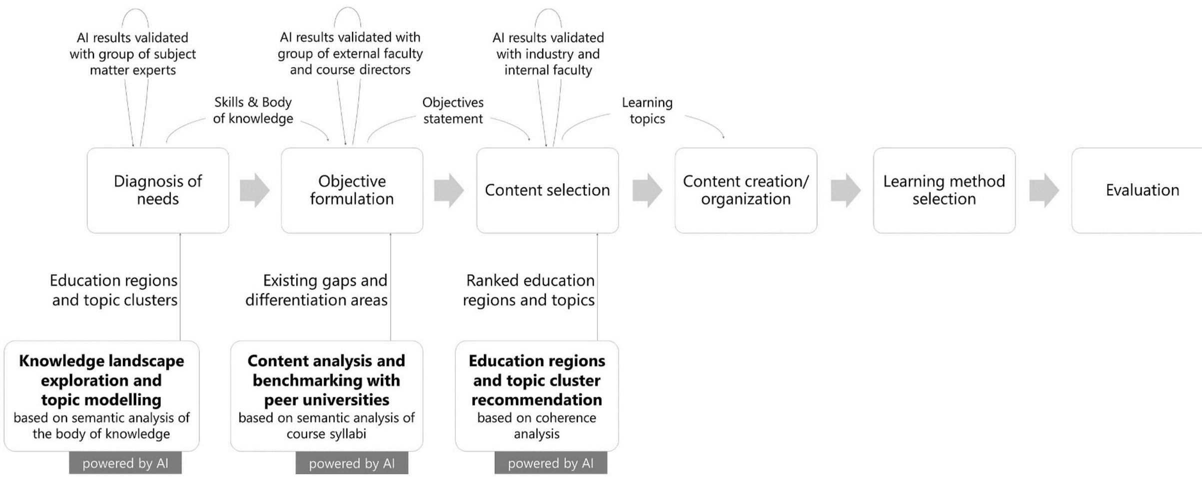

Fig. 2. AI-powered curriculum development framework.

a nuanced understanding of the multifaceted industry landscape. Results are presented in Sec. 4.1.

3.3. Curriculum positioning through benchmarking and diversification

Positioning involves strategically situating the curriculum (or program or educational portfolio) in the national (and international) educational landscape to maximize student attractiveness. Research findings show that diversification from their main national and international competitors is beneficial to universities in attracting more freshers (Biagi et al., 2024), in particular for peripheral and small regions that persistently experience outbound student migration flows. This delicate endeavor balances continuity and innovation in education, aligning with existing curricula while infusing it with a unique competitive edge. It demands a distinctive character that attracts and retains students, embracing innovative content to reflect evolving industry dynamics and technological advancements. Traditionally, this process required manual review of existing curricula, identifying areas of congruence and divergence, and crafting strategies for differentiation.

To streamline the complexities of curriculum alignment, an AI-based tool was introduced in this research. This tool performs extensive semantic analysis of curriculum syllabi, enhancing the traditionally manual and time-consuming process of identifying congruence and divergence between proposed and existing curricula. By leveraging NLP and ML algorithms, the tool adeptly evaluates curriculum content, offering real-time insights for differentiation and fostering more agile and responsive curriculum development.

3.3.1. Benchmarking with existing curricula

In this study, a comprehensive exploration of IEM degree programs in Italy was conducted. Data was sourced at the beginning of 2023 from official program sheets (the SUA-CDS, in Italian: Scheda Unica Annuale) via the Universitaly website,[2] the platform administered by the Italian Ministry of University and Research dedicated to cataloging data on bachelor and master's programs. At the end of the selection process, 83 courses served as the basis for further analysis. First, the developed tool employs data preprocessing techniques to enhance the quality and consistency of the syllabi, encompassing measures such as the removal of punctuation, stop words, and text normalization through stemming and lemmatization. The text was tokenized by word and transformed lowercase to avoid case-sensitive bias. Furthermore, words shorter than 4 letters were omitted to avoid considering acronyms. Second, we leveraged a multilingual SBERT model (similar to the one above) to map sentences and paragraphs to a multidimensional vector space thanks to a pre-trained deep network and then mean pooling is applied to represent documents in the space. The tool clustered the syllabi (expressed as points in a multidimensional vector space) according to the k-Means clustering algorithm. The tool tests different numbers of clusters and calculates the clustering scores for the selected cluster range using the silhouette score. Silhouette analysis can be used to study the separation distance between the resulting clusters. k-Means++ has been used as initialization method with 10 re-runs and 300 maximum iterations. Also in this case, to visualize the curricula in an interpretable way, data dimensionality was reduced from 384 to 2 using t-SNE visualization. After calculating the t-SNE map, to develop a semantic annotation scheme for the clusters identified through the k-Means algorithm, cluster keywords are extracted with the TF-IDF (Term Frequency-Inverse Document Frequency) method and most frequent bigrams and trigrams in the corpus were calculated. These two results will be used by the curriculum developers as relevant input to position the new programme in a way to be relevant but also different than other existing courses, as explained in the Results section.

3.3.2. Benchmarking and lessons learned from Erasmus + projects

This benchmarking analysis is enhanced through an exploration of projects funded under key action 2 of the Erasmus+ program. Such action supports transnational cooperation to modernize and strengthen educational offers at higher education institutions. Through the examination of these projects, we gained insights into how other institutions are addressing similar educational challenges and which topics they are focusing on. The Erasmus + project results platform[3] was used to screen and select relevant projects. In particular, we filtered out the projects by: i) *keywords* ("industry 4.0", "industry 5.0" OR factory OR manufacturing); ii) *education level* (only projects whose action type falls in the higher education level were considered); iii) *availability of project results* (only projects with available results on the platform were considered); iv) *topic* (only projects falling under the topics "New innovative curricula/educational methods/development of training courses" were considered). 26 projects resulted from the project search, but eventually only 18 projects (see in Annex A) were considered as the scope of the other 8 projects did not match the one of this study. Project cards and reports available on the project results platform served as basis for further research through an approach similar to the one described above (cf. Sec. 3.3.1).

[2] https://www.universitaly.it/.

[3] https://erasmus-plus.ec.europa.eu/projects.

3.4. Educational areas and learning topics recommendation

The selection and recommendation of educational areas and learning topics can be informed by the results provided by the AI-based tool. We employ here a quadrant-based approach, which categorizes educational areas based on two fundamental parameters: the maturity level in the scientific knowledge landscape extracted during the first step (Sec. 3.2) and the number of existing courses extracted in the second step (Sec. 3.3.1). The framework classifies educational areas into four distinct quadrants, each representing a unique set of opportunities and challenges for curriculum designers. The graph is obtained by plotting all the educational areas and using their normalized weights (cf. Table 2) as x-axis and the percentage of existing courses mentioning that area in their syllabi (cf. Table 4) as y-axis. The matrix will represent a user-friendly tool informed by AI that curriculum developers can use to make their final decision.

3.5. Delphi validation process of AI-powered results

A critical facet lies in the validation of curriculum recommendations generated by the AI-based tool. Through multiple survey rounds, the Delphi validation process has been conducted to harness the collective wisdom of panel of recognized experts within the IEM field, carefully selected for their expertise and knowledge, to evaluate and rate the relevance and importance of AI-generated curriculum recommendations. Experts were interviewed during the roundtable on "Industrial Engineering 4.0 Education Initiative" that was held in Linz (Austria) on November 4th, 2022 at the 4th International Conference on Industry 4.0 and Smart Manufacturing (ISM 2022). The roundtable included 15 experts and professors of Industry 4.0 and Smart Manufacturing attending the conference from different countries. At the beginning of the session, informed consent was obtained from all participants, and their privacy rights were strictly observed. They knew that the participation was voluntary and they could withdraw from the study at any time. Questionnaires were anonymous and during the discussion anonymity was respected. The participants were protected by hiding their personal information during the research process. In the first round, a questionnaire was distributed to the experts via Google Form. The questionnaire contained the list of 24 educational areas resulting from Step 1 and

Table 2
Education regions and weight.

Region		Weight
#	Name	
1	Technology and innovation management	12.24%
2	Adaptable and reconfigurable manufacturing systems and processes	5.34%
3	Internet of things, networking and communication	7.08%
4	Simulation and digital twins	2.36%
5	Extended reality	1.44%
6	Interoperability and system integration	3.35%
7	Big data, information engineering and analytics	5.16%
8	Artificial intelligence for manufacturing	5.44%
9	Cyber-physical systems and control systems	5.38%
10	Robotics and self-X systems	2.16%
11	Decentralized systems and cybersecurity	3.37%
12	Human-machine interface engineering	1.73%
13	Production optimization and analytics	4.97%
14	Manufacturing process digitalization and innovative techniques	4.10%
15	Maintenance, quality and reliability engineering	5.47%
16	Production intelligence and data-driven process monitoring	2.97%
17	Business economics and transition to economy 4.0	3.00%
18	System and product design, modeling and engineering	2.42%
19	Logistics and supply chain management	2.28%
20	Safety engineering and management	1.45%
21	Society and organization management	5.36%
22	Sustainability and environment	4.72%
23	Policy and standards	1.76%
24	Case studies	6.44%

Table 3
Education regions and topic clusters (clusters with a weight smaller than 3% are not mentioned in this table for a matter of space).

Region		Clusters (and in-cluster weight)
#	Name	
1	Technology and innovation management	Industrial transformation and intelligent systems (19.9%); digitalization and digital transformation (19.8%); technology landscape and emerging trends (12.9%); maturity models and adoption frameworks (10.0%); automation and intelligence (9.6%); product-service systems, service-oriented manufacturing and manufacturing-*as*-a-service (6.1%); innovation management (5.3%); business model innovation (4.4%); personalization and mass customization (3.5%); reference models and frameworks (3.2%); change management, usability and acceptance (3.1%); resilience (3.0%)
2	Adaptable and reconfigurable manufacturing systems and processes	Production systems and process theory (50.6%); manufacturing scheduling (16.6%); assembly systems (13.8%); reconfigurability, flexibility, modularity and adaptability (9.5%); shop floor management (4.5%); cellular manufacturing, production cells (3.0%)
3	Internet of things, networking and communication	Internet of things (44.4%); sensors (10.1%); networking and connectivity (10.1%); SCADA (7.2%); M2M communication (5.5%); wireless sensor networks (4.8%); wearable and mobile devices (3.8%); 5G, mobile and internet communication (3.7%); Radio Frequency Identification, RFID (2.7%)
4	Simulation and digital twins	Digital twins (31.7%); modeling and model-based engineering (24.2%); simulation (23.4%); multi-agent modeling (13.4%); discrete-event simulation (7.2%)
5	Extended reality	Virtual reality (44.4%); augmented and mixed reality (38.6%); 3D modeling (9.3%); head-mounted displays (3.9%); gamification (3.2%)
6	Interoperability and system integration	Software applications and middleware (18.6%); interoperability and system integration (17.5%); computer architecture, hardware/software paradigms (16.7%); semantic modeling and interoperability (10.9%); ontology-driven engineering (6.1%); OPC UA (5.2%); web services and web-based applications (5.2%); closed-Loop control systems and end-to-end integration (4.7%); open systems and open source software (4.5%); asset administration shell (4.2%); data issues (3.4%); IT/OT convergence and synchronization (3.0%)
7	Big data, information engineering and analytics	Data analytics and integration (35.0%); cloud computing and cloud-based manufacturing (17.5%); big data (14.5%); knowledge management and representation (10.6%); forecasting and predictive analytics (7.8%); time series analysis (6.0%); edge computing (4.4%)
8	Artificial intelligence for manufacturing	Machine learning (30.4%); artificial intelligence in manufacturing (17.1%); computer vision (10.2%); neural networks (8.2%); data mining (6.4%); image processing and classification (4.5%); pattern recognition and data fusion (3.7%); feature extraction and recognition (3.3%); information classification methods (3.3%); natural language processing (3.3%); transfer

(*continued on next page*)

Table 3 (*continued*)

Region #	Name	Clusters (and in-cluster weight)
		learning (3.1%); object detection and recognition (3.0%)
9	Mechatronics and cyber-physical systems	Cyber-physical systems (41.1%); embedded systems (23.0%); control systems and industrial automation (13.0%); real-time systems monitoring and control (11.6%); programmable logic controllers (3.1); microelectronics and controllers (3.1%); mechatronics (3.0%)
10	Robotics and self-X systems	Industrial Robots and Collaborative Human-Robot Interaction (75%); manipulators and gesture recognition (5.2%); Self-X systems (4.4%); autonomy and autonomous systems (4.4%); orchestration (3.1%)
11	Decentralized systems and cybersecurity	Information management (36.4%); data privacy and cybersecurity (34.3%); distributed computer and decentralized control systems (12.1%); blockchain (9.2%); digital storage (8.0%)
12	Human-machine interface engineering	Human-centric manufacturing and human factors (29.6%); human-machine interaction (17.3%); exoskeletons (10.6%); operator 4.0 (9.1%); haptic interfaces and perception (6.9%); human-machine systems (6.8%); graphical user interfaces (5.9%); coordination and symbiosis (4.9%); chatbots and speech recognition (4.5%); interaction strategies and mechanisms (4.2%)
13	Production optimization and analytics	Decision making and decision support systems (26.1%); optimization techniques (25.8%); analytical and statistical models (7.8%); stochastic systems and fuzzy modeling (5.0%); decision processes and clustering analysis (4.3%); multivariate, multi-criteria and multi-objective analysis (3.3%); uncertainty analysis and probabilistic models (3.3%)
14	Manufacturing process digitalization and innovative techniques	Additive manufacturing and 3D printing (23.0%); machine tools and CNC (20.4%); flow control and optimization (10.5%); machining and material processing techniques (8.8%); cutting, casting and welding (6.7%); finite element modeling (6.2%); heat transfer and thermal control (6.0%); vibration analysis, radio wave and ultrasonic applications (4.1%)
15	Maintenance, quality and reliability engineering	Predictive and smart maintenance (19.7%); quality control and management (16.0%); Six Sigma and Lean (15.2%); anomaly and defect detection and prediction (13.7%); condition monitoring and preventive maintenance (11.0%); prognostics and health management (5.1%), system reliability analysis (4.7%); inspection and quality assurance (4.1%), traceability, detection and tracking (3.3%)
16	Production intelligence and data-driven process monitoring	Process modeling and monitoring (34.3%); performance measurement systems and management (30.9%); business intelligence (17.7%); value creation and value chain (11.9%); computer programming and low-code/no-code programming (3.3%); continuous improvement and kaizen (3.0%)
17	Business economics and transition to economy 4.0	sales, customer-centric strategies and mass customization (25.2%); cost benefit analysis and cost engineering (24.8%); economic profitability and success factors

Table 3 (*continued*)

Region #	Name	Clusters (and in-cluster weight)
		(11.8%); global competition and sustainable competitiveness (16.5%); investments, budget control and finance (6.9%)
18	System and product design, modeling and engineering	Conceptual and computer-aided engineering design (22.5%); product development process (21.4%); technical design (11.2%); functional analysis and design (10.5%); structural modeling (8.8%); conceptual and formal methods (7.7%); testing (6.2%); complexity and complex Systems engineering (4.4%); constraints and barriers (4.2%); reverse engineering (3.0%)
19	Logistics and supply chain management	Supply chain management (46.8%); internal logistics and warehouse automation (18.4%); autonomous vehicles, AGVs and intelligent transportation systems (12.5%); materials handling (10.3%); spare parts management (8.0%); transportation systems, railroads, and sustainable Mobility (4.2%)
20	Safety engineering and management	Occupational health and safety (22.9%); physical and cognitive ergonomics (21.4%); risk management (17.6%); safety management (16.8%); accident and disaster prevention and management (14.4%); stress and fatigue (3.9%)
21	Society and organization management	education and pedagogy (20.1%); engineering education (12.3%); employment, skills and jobs (9.9%); human resource management (9.0%); management systems and leadership (6.9%); agile manufacturing systems (6.3%); work environment, work stations, workplace, work simplification, and workflow management (5.7%); learning factories and industry academia collaboration (4.4%); personnel training (4.1%); social aspects and socio-technical systems theory (3.7%); organization management: structure, culture and readiness (3.6%)
22	Sustainability and environment	Sustainable development (32.4%); energy efficiency and management (16.5%); life cycle management (14.4%); environmental impact, protection and management (6.6%); circular economy (6.4%); carbon footprint and pollution (5.5%); natural resource management (3.8%); green manufacturing (3.3%); waste reduction and management (3.0%)
23	Policy and standards	Country-based studies (34.4%), standards and regulatory framework (31.7%), international trade and globalization (15.9%), best practices (8.9%), policy analysis and implications (6.2%)
24	Case studies	Global and multinational companies (7.5%), small and medium enterprises (6.8%), steelmaking and metal industries (6.5%), automotive industry (6.3%), case-based reasoning (4.6%), agriculture and smart farming (4.2%), electronics manufacturing and circuit design (3.7%), power generation and consumption (4.1%), electronics manufacturing and circuit design (3.7%), construction industry (3.3%), aerospace industry: drones, aircraft, and unmanned aerial vehicles (3.1%), pandemics (3.1%), optical technologies in glass and light-related industries (3.0%)

Table 4
Most frequent bigrams and trigrams in the corpus. Relevant n-grams are in bold.

student able, case study	>0.20%
digital twin, supply chain, course student, end course	0.15%–0.20%
decision making, internet thing, manufacturing system, smart grid, student learn, **artificial intelligence**, course aim, **iot system**	0.10%–0.15%
life cycle, technological change, machine learning, course provide, **industrial asset, production system**, student acquire, **simulation model, acquired knowledge, data integration, enabling technology, industrial plant**, performance cost, provide student, research development, **asset management, digital technology, energy system**, aim course, aim provide, **big data, data analysis, data science, impact technological, industrial iot, knowledge understanding**, module student, new technology, objective course, recorded lecture, **service design**, specific problem, **value innovation, work organization**, allow student, **business model, collaborative robot, communication technology**, industrial environment, **sensor network**, able apply, able understand, apply acquired, complex system, **cost risk**, discrete event, exercise case, **information system, integrated manufacturing**, method tool, physical asset, problem setting, **product life, shop floor, space economy**	0.05%–0.10%
business industrial, course cover, **data processing, design solution, digital model**, following topic, industrial application, learning goal, **maintenance management, measurement system, product service, real time, security management, smart factory**, study session, system student, **technological value, asset lifecycle, communication skill, data analytics**, design experiment, **displacement velocity**, e activity, **event simulation**, floor management, give student, hour theory, **industrial maintenance**, particular course, production planning, **smart maintenance, spatial data**, theory matlab, use case, application domain, **collaborative robotics, communication protocol**, context function, course also, **digital shop, digital transformation, dynamic model**, dynamical system, **emerging technology**, end module, **ethical social**, function process, goal course, hour student, introduce student, know understand, **law technology**, learning outcome, lecture given, load cell, making process, new business, part course, physical system, project work, real case, sensor node, system industrial, technology able, theoretical lesson, topic covered, ability apply, **application layer**, comprises hour, course includes, course provides, **data acquisition**, delivery lecture, dimensional digital, distribution system, effort required, **environmental monitoring**, given time	0.03%–0.05%

experts were asked to rate their importance (how important is the educational area to train an expert in industry 4.0 or 5.0?) and their popularity in educational systems ('how many courses exist according to your knowledge?') on a scale 1 to 5, with 1 being "not important" and 5 being "very important". We analyzed then the responses to identify the topics that received a high level of consensus and reviewed the comments provided by experts. In the second round, results were shared with experts, who were asked to re-evaluate the topics that did not reach a clear consensus in the first round. In the final round, we presented the list of curriculum recommendations based on the feedback and revisions made in the second round. Results include topics that achieved a high level of consensus and those that remained contentious. Such topics were compared to the one provided by the AI tool.

4. Results

4.1. Knowledge landscape and educational regions

The semantic analysis of the knowledge landscape yielded a dataset of 19,744 keywords, which were then visualized using the t-SNE map depicted in Fig. 3. In this bidimensional representation, every data point is a keyword. The size of each data point, represented as a bubble, corresponds to the frequency of occurrence of the keyword as observed in the literature. The proximity of data points is an indication of the semantic similarity between the two of them, with shorter distances indicating higher similarity between keywords. Notably, the spatial arrangement in this two-dimensional space maintains a similarity to the original 384-dimensional space, thereby preserving the underlying data structure. This process effectively grouped related keywords into 357 distinct clusters, referred to as 'learning items' or 'topics', that are the elementary components in our curriculum design. In the t-SNE map in Fig. 3, the bubbles are color-coded to denote their assigned clusters, which were established through hierarchical clustering utilizing the Ward's algorithm. Some sample cluster labels are shown in Fig. 3 for a matter of understandability. The complete interactive graph can be conveniently accessed online at https://public.flourish.studio/visualisation/15211024/.

Subsequently, these 'topics' were aggregated into broader 'education regions', each embodying high-level subjects that constitute the 'industry of the future' knowledge landscape. Such regions might be considered as the modules of our curriculum. For a comprehensive overview of these areas, please refer to Table 2, which enumerates the assigned names for the 24 regions. For a matter of space, Table 3 details the most salient topics, identified by a minimum weight threshold of 3.0%, within each area. The full dataset is also available online and open access.[4] This list of topics serves as a foundational guide for ensuing curriculum design and development stages.

4.2. Benchmarking analysis for diversification of the educational offer

The benchmarking analysis generates a scatter plot visualization, known as a t-SNE map, which is depicted in the left section of Fig. 4 a. The map reveals the spatial arrangement of course syllabi and serves to identify clusters of courses with semantic relatedness. Additionally, our implemented tool demarcates these clusters with hulls, as illustrated in Fig. 4 b, while simultaneously providing a collection of keywords associated with each cluster. To assess the quality of clustering, we employed silhouette analysis. Silhouette plots offer a quantitative measure of the proximity of each data point within a cluster to the neighboring clusters. They provide a visual method to evaluate clustering parameters, such as the number of clusters. Silhouette coefficients near +1 suggest that the data points are distinctly separated from neighboring clusters. A value of 0 implies that a data point is situated near or on the boundary between two adjacent clusters, while negative values indicate potential misallocation to clusters. A comprehensive evaluation in Fig. 4 c reveals that the average silhouette score for each instance within its respective cluster consistently surpasses the 0.5 threshold. This achievement signifies the effectiveness of our clustering tool, particularly considering the relatively limited dataset comprising only 83 instances, with syllabi referencing a multitude of topics. The tool further provides users with a table featuring the most frequent bigrams and trigrams within the corpus. Among these, the most important ones are denoted in bold in Table 4. Curriculum developers can effectively utilize these outcomes as valuable inputs to position the new program in a manner that aligns with relevance while distinguishing it from existing courses, as elaborated in the ensuing sections. It's essential to acknowledge that this method relies on the assumption that curriculum syllabi have been meticulously prepared and accurately transferred to the Universitaly platform by educators. However, a limitation of this approach stems from the possibility that syllabi may not always comprehensively reflect the actual course content and could potentially contain inaccuracies.

4.3. Insights from Erasmus + projects

From the analysis of the Erasmus + projects, it emerged that only very few projects provide a holistic curriculum for industry 4.0 experts

[4] Full dataset: Padovano, Antonio (2023), "Curriculum development for engineering education 5.0: a text mining and natural language processing tool", Mendeley Data, V1, doi: 10.17632/k98fm2jdg7.1.

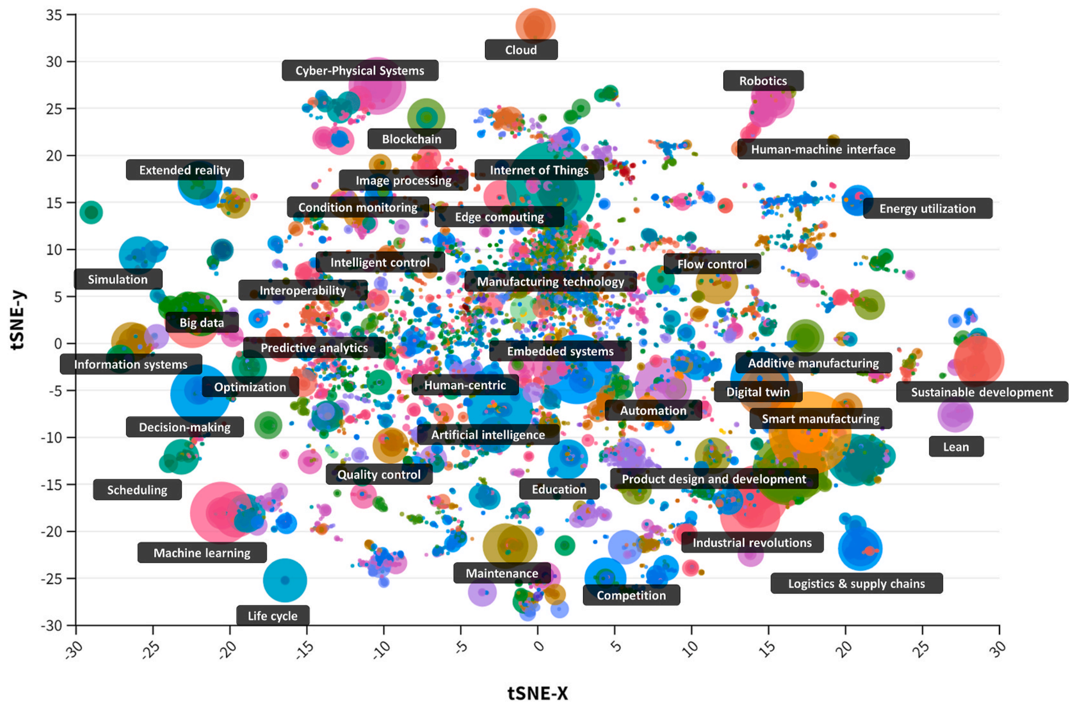

Fig. 3. Clustering keywords in the Industry 4.0 and 5.0 corpus. Sample cluster labels are shown. Full interactable graph accessible at https://public.flourish.studio/visualisation/15211024/.

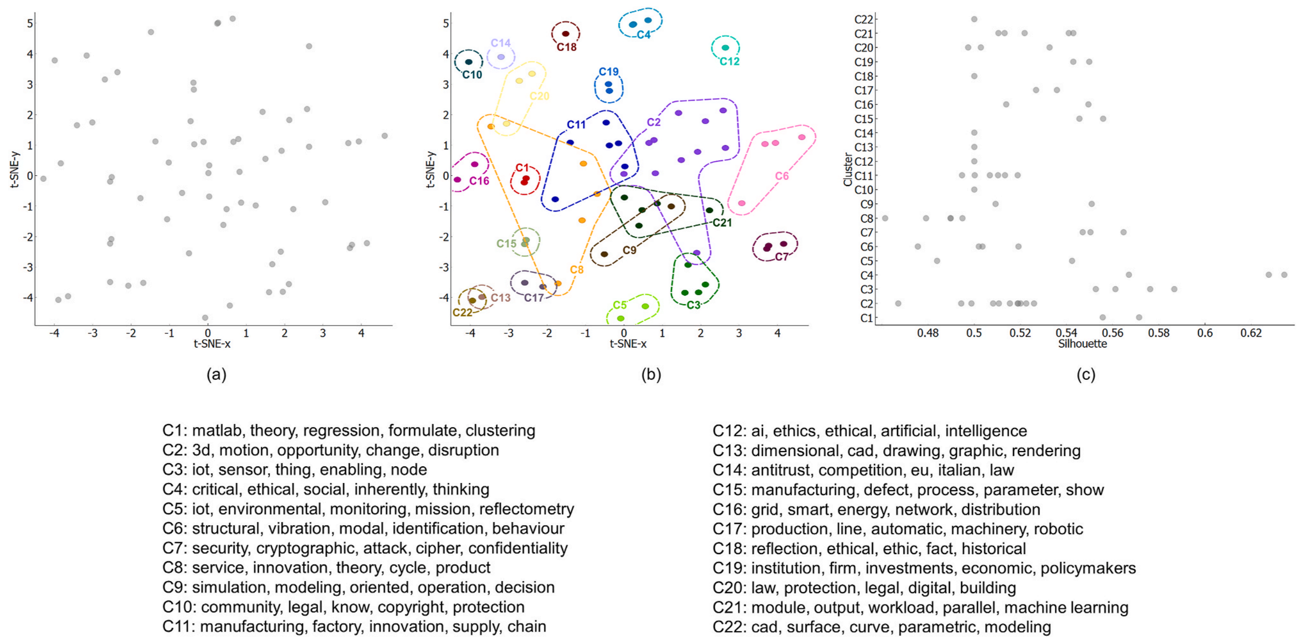

Fig. 4. Clustering university courses by topic.

(2019-1-MT01-KA203-051265). Most projects aims at developing new curricula in thematic areas of Industry 4.0. The majority focused on the basics of mechatronics systems, industrial automation, Information Technology/Operational Technology (IT/OT) convergence and industrial IoT (2019-1-RO01-KA203-063153, 2020-1-ES01-KA226-HE-095291, 2017-1-IT02-KA203-036980, 2016-1-ES01-KA203-024921). Some projects focused instead on manufacturing system performance analysis, methods and time measurements and ergonomics, also via virtual and augmented reality (2018-1-EE01-KA203-047094) or digital and non-conventional manufacturing, e.g. additive manufacturing (2019-1-RO01-KA203-063486). Strictly connected to the additive manufacturing projects, a few of them focused on the product development process and rapid prototyping (2018-1-TR01-KA203-059739, 2017-1-SI01-KA203-035522). Surprisingly, only few projects focus on robotics in industry 4.0 (2019-1-ES01-KA203-065796, 2019-1-MT01-KA203-051265). A small set of projects focused instead on how industry 4.0 technologies are revolutionizing some areas, such as purchasing and procurement (2019-1-NL01-KA203-060501) and logistics and supply chain (2019-1-PL01-KA203-065731, 2017-1-PL01-KA203-038698) with the aim to understand and train the next generation of purchasers 4.0 or logistics manager 4.0. Instead, only one project explicitly deals with social aspects of industry 4.0 and corporate social responsibility (2020-1-BG01-KA203-079025).

To summarize, Table 5 reports a score (0–4) associated to each education region identified in the previous section. The score is a qualitative indicator of how many ERASMUS + projects focused on that specific area: 4 – the region appears in the majority of projects; 3 – the region is a recurring theme; 2 – the region appears in some projects, but it is not recurring; 1 – the region sporadically appears; 0 – the region never appears in the projects. This analysis served as an interesting element for the selection and recommendations of educational areas. This analysis provides an important recommendation to educators and practitioners who wish to define novel curricula as well as to the policy makers and European commission that can use this table as a guideline on which education areas require more development and attention (e.g. in ERASMUS + projects). Areas with only one point (●) or less are those without a clear curriculum or educational material and that have received poor attention from the community. Therefore, they might be of greatest interest for the development of new curricula and competitive positioning of a higher education institution to attract students. We understand that one limitation is that only ERASMUS + projects are considered here. In the future, other actions (e.g. Horizon projects might be incorporated in the analysis) can be considered.

4.4. Educational area and topic recommendation

The resultant quadrant-based matrix is visually depicted in Fig. 5, wherein four distinct quadrant are discernible, each representing a unique strategy.

- **Quadrant I - Innovate and Pioneer**: In this quadrant, topics are still in the early stages of scientific development, and few existing courses are available. Curriculum designers in this quadrant are encouraged to adopt an innovative and pioneering approach. They have the opportunity to introduce entirely new courses or educational pathways that align with emerging fields of knowledge. Quadrant I encompasses fertile ground for pioneering curricula, exemplified by areas such as 'interoperability and system integration', 'simulation and digital twins', 'extended reality', 'human-machine interface engineering', 'decentralized systems and cybersecurity', and others. Notably, this quadrant exhibits the highest population, underscoring the prevalence of underdeveloped areas that necessitate inclusion in structured Industry 4.0 or 5.0 educational programs.
- **Quadrant II - Specialize and Develop**: Quadrant II includes areas featuring topics with high scientific maturity but a limited number of associated curricula. Here, there is substantial potential for specialization and curriculum development. Designers are encouraged to craft specialized tracks and advanced courses that comprehensively cover these mature topics. Some examples in this quadrant include 'artificial intelligence for manufacturing', 'case study analysis', 'maintenance, quality, and reliability engineering,' and 'adaptable and reconfigurable manufacturing systems'.
- **Quadrant III - Monitor and Expand**: Quadrant III encompasses educational domains that exhibit a high level of scientific maturity, accompanied by a significant number of existing courses. The size of data points within this quadrant reflects the availability of curriculum materials, as sourced from Erasmus + projects. In this quadrant, curriculum developers have an opportunity to explore these well-established areas of knowledge. They can seek to enhance existing courses with the latest research findings. Notable topics in Quadrant III comprise 'technology and innovation management', 'internet of things, networking and communication', 'mechatronics and cyber-physical systems', and 'society and organization management'.
- **Quadrant IV - Explore and Enhance**: Quadrant IV pertains to educational areas characterized by relatively new topics in the scientific landscape that already boast a substantial number of existing courses. Here, the emphasis lies in monitoring emerging trends and expanding existing curricula to meet the growing interest and demand. This quadrant includes areas like 'big data, information engineering and analytics' (positioned at the center of the map), 'system and product design, modeling, and engineering', 'logistics and supply chain management', and 'policy and standards'.

Table 5
Presence of the educational area in completed ERASMUS + projects.

Region #	Region Name	Connections with ERASMUS + projects
1	Technology and innovation management	●
2	Adaptable and reconfigurable manufacturing systems and processes	
3	Internet of things, networking and communication	●●●●
4	Simulation and digital twins	●
5	Extended reality	●●
6	Interoperability and system integration	●●●
7	Big data, information engineering and analytics	●●●
8	Artificial intelligence for manufacturing	●
9	Mechatronics and cyber-physical systems	●●●
10	Robotics and self-X systems	●
11	Decentralized systems and cybersecurity	●
12	Human-machine interface engineering	
13	Production optimization and analytics	
14	Manufacturing process digitalization and innovative techniques	●●
15	Maintenance, quality and reliability engineering	
16	Production intelligence and data-driven process monitoring	
17	Business economics and transition to economy 4.0	●
18	System and product design, modeling and engineering	●●●
19	Logistics and supply chain management	●●●
20	Safety engineering and management	
21	Society and organization management	●
22	Sustainability and environment	●
23	Policy and standards	●
24	Case studies	●●

Quadrant I and II highlight areas of interest for innovative curricula, whereas Quadrant III identifies topics worthy of consideration for alignment with educational offerings. On the other hand, Quadrant IV represents areas that may not be preferable, either due to the underdevelopment of the field in the context of Industry 4.0 or 5.0, or because alternative courses already adequately address the subject matter.

4.5. Validation

The validation process revealed compelling insights into the consensus and areas of contention. Notably, the Delphi process underscored the unanimous agreement among experts on the paramount importance of 'human-centric IEM' as a focal area within the curriculum. For example, many experts mentioned the relevant role of ethics (54% of experts), socio-technical systems (27%) and socio-economic aspects (34%) in innovative curricula. The convergence of expert opinions on this topic emphasized its significance in preparing future IEM students. This is in line with the Industry 5.0 paradigm that identifies human-centricity as one of the pillars of the next industrial revolution. Furthermore, 'digital twin engineering' emerged as another strongly agreed-upon area, reflecting the growing relevance of digital twin technologies in modern industrial settings. However, the Delphi process also illuminated areas of conflicting opinions among experts. For instance, divergent views were prevalent when considering the inclusion of 'blockchain applications in industry 4.0'. While some experts advocated for its inclusion as a forward-looking topic in line with Industry

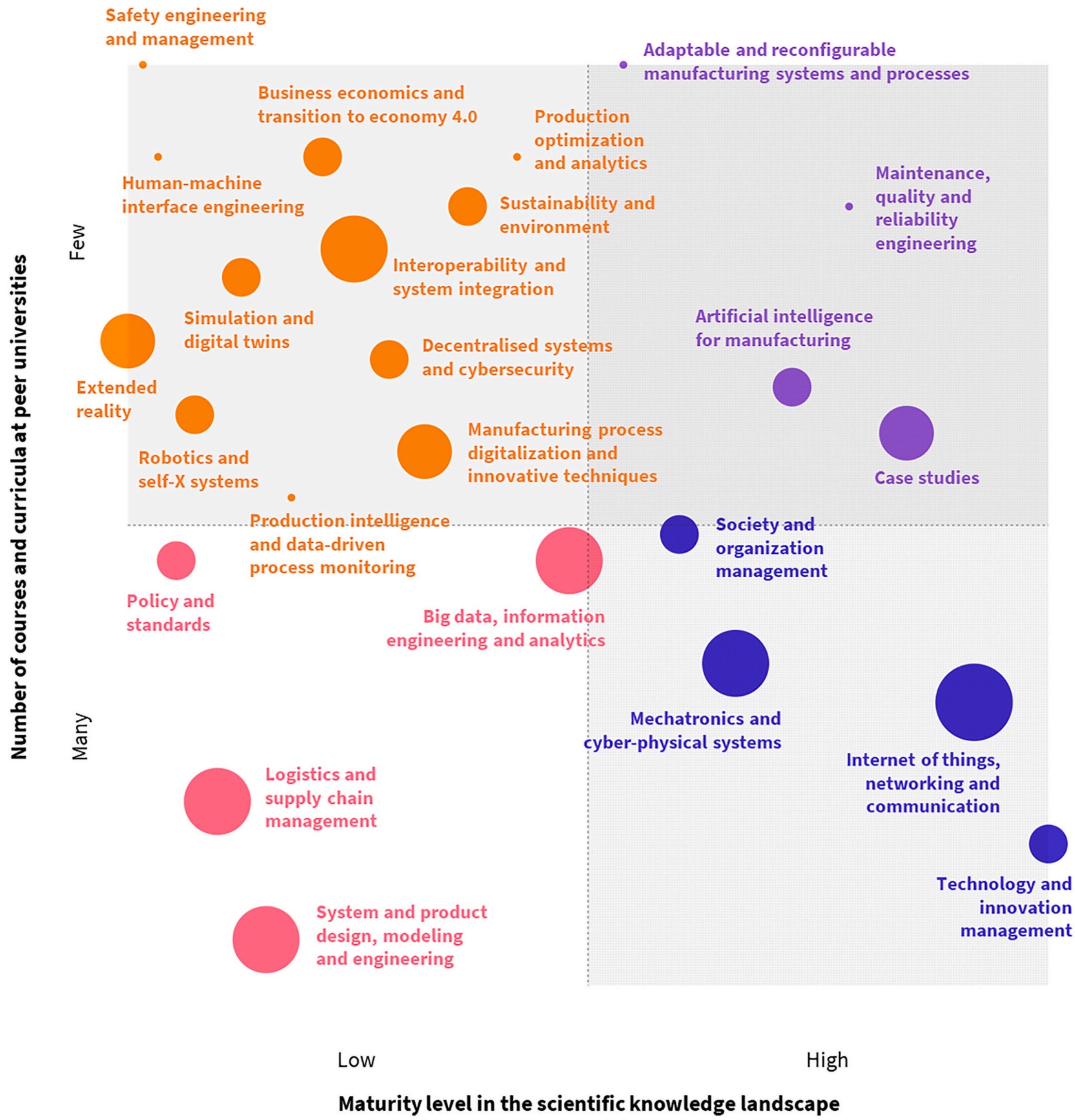

Fig. 5. Quadrant-based matrix for educational area selection.

4.0, others expressed reservations, questioning the maturity and practicality of teaching blockchain applications for IEM students. In the realm of ’circular economy principles in industrial processes’, mixed opinions were evident. While all experts championed its incorporation as a pivotal sustainability aspect, some of them raised concerns about the interest as new and innovative course considering that many HEIs have already effectively integrated this subject into courses. Importantly, the expert consensus identified in the Delphi process is found to be in alignment with the outcomes generated by our AI-based curriculum analysis tool. Experts largely agreed on the importance of introducing pioneering courses and educational paths to align with the emerging fields of knowledge classified in Quadrant I. However, there were also areas where conflicting viewpoints emerged. For instance, certain experts advocated for further emphasis on topics in Quadrant I, suggesting the need for more pioneering courses, while others argued for Quadrant III areas, emphasizing the enhancement of existing curricula. These divergent perspectives highlight the complexity and dynamic nature of curriculum development in the context of Industry 4.0 and 5.0 calling for human decision at this time of the process. However, the convergence between the results achieved through AI and through the Delphi process serves as a robust validation of the AI-driven approach, reinforcing the credibility and relevance of the proposed tool.

5. Discussion

Using AI and data-driven approaches to explore the knowledge landscape and support the curriculum development process offers several advantages over relying solely on human experts. In particular, AI provides systematic approach to analyzing vast amounts of information, which can be especially beneficial when dealing with complex and rapidly evolving domains like the ’industry of the future’. The tool is in fact scalable. AI can efficiently process and analyze large volumes of data, making it possible to consider a wide range of knowledge sources and areas (like we did in this study, e.g. SCOPUS metadata, course syllabi, Erasmus + projects results), which may not be feasible for a human expert to handle comprehensively. Also, AI algorithms do not have inherent biases, ensuring an objective and consistent evaluation of information. Human experts may unintentionally introduce personal biases or be influenced by preconceived notions. Most importantly, AI can continuously monitor and analyze new data, ensuring that recommendations and insights are up-to-date and aligned with the latest developments, which can be challenging for human experts, deans, program directors to maintain. Last but not least, the developed approach can significantly expedite the analysis process, potentially reducing the time and cost associated with manual expertise and research. Overall, employing AI, NLP and text mining to explore the knowledge landscape complements human expertise and enhances the depth, breadth, and efficiency of the analysis, making it a valuable tool for curriculum development and decision-making.

5.1. Towards a human-AI collaboration in competency-based curriculum design

AI-driven CBE facilitates a culture of continuous improvement and adaptation within HEIs and presents transformative benefits, as evident from various studies. ML and text mining methods are beginning to play a substantial role in curriculum development (Tavakoli et al., 2022). In particular, Bahroun et al. (2023) identify AI-enhanced curriculum design as one of the most promising future research directions in educational research. Our findings are in line with previous studies that reported the benefits of using AI for analyzing curricula against textbooks (Yang et al., 2023). At the same time, while this study undoubtedly showcases the positive implications of AI for streamlining the curriculum design process, it also unveils crucial issues that demand careful considerations, for example an overreliance on AI results or recommendations. While this research accentuates the transformative potential of AI in revolutionizing the educational landscape, we underscore the urgency for judicious implementation and the need of a synergistic human-AI collaboration in the competency-based curriculum design process. In this paper, we argue that AI serves as a valuable ally (or *curriculum design assistant*) capable of analyzing multimodal data and recommending the optimal strategies for selecting learning content or positioning curricula within the educational landscape. This study aims to inspire researchers in education science and AI to focus on how the curriculum design process (see Figs. 1 and 2) can be further supported by AI and data-driven insights. We argue that HEIs should open to a more structured and *collaborative continuous development* of CBC. By automating the analysis of large datasets, AI algorithms can quickly identify patterns and trends, allowing HEIs to develop and update competency-based curricula in a more timely and cost-effective manner. This enables HEIs to adapt their programs more rapidly to changing industry needs and technological advancements.

5.2. Generative AI in competency-based curriculum design

Generative AI is revolutionizing education by going beyond traditional AI applications that focus on data analysis, assessment automation, and personalized learning recommendations. Unlike conventional AI, which suggests existing content based on student data, generative AI creates new instructional materials tailored to individual learning needs, ensuring that resources are current and aligned with the latest standards and industry demands. This capability supports dynamic content creation and adaptive learning pathways, allowing real-time adjustments to educational materials based on student performance and providing more relevant and challenging tasks. Additionally, generative AI facilitates rapid prototyping and iterative curriculum design by helping educators develop and refine new educational content efficiently (cf. Step 4 in Figs. 1 and 2). This potential is particularly significant given the immense effort required from educators, teachers, and professors to keep educational resources up-to-date amidst their already demanding schedules. Generative AI can significantly alleviate this burden by dynamically creating and updating educational content, allowing educators to focus more on teaching and less on content creation. It also promotes collaborative content creation, generating initial drafts that educators can enhance collectively. However, the use of generative AI must address ethical challenges, such as mitigating algorithmic biases to ensure fair educational outcomes and implementing robust measures to protect learner data privacy and security. The synergy between educators and generative AI holds immense promise for a more responsive and personalized educational experience while maintaining ethical integrity.

5.3. Data quality and study assumptions

The application of AI tools in curriculum development, as exemplified in this study, brings to the forefront important considerations regarding data quality. One notable challenge lies in the accuracy and completeness of the underlying data. Inaccuracies or gaps within the dataset utilized for semantic analysis can exert a profound influence on the efficacy of AI-driven tools. If the initial dataset contains errors, inconsistencies, or outdated information, it can potentially yield flawed insights and recommendations.

First, our findings echo the conclusions drawn by Mejia et al. (2021). We underscore the critical reliance of semantic approaches on trustworthy ground truth datasets. Our study aligns with this perspective, emphasizing the necessity of reliable input data to construct pertinent and discipline-specific knowledge landscapes. By employing reproducible methods, our research contributes to the informetrics field while proposing a complementary solution. We advocate for further exploration in this area, emphasizing the importance of utilizing trustworthy datasets as foundational elements for future studies.

Second, it is worth noting that data contained in syllabi, often the basis for such analyses, can suffer from obsolescence due to the dynamic nature of educational content. This study operates under the premise that the syllabi procured from sources like the Universitaly portal are both accurate and faithful representations of course content. This assumption carries implications for educators and institutions alike. It underscores the necessity for educational providers to maintain up-to-date documentation of their courses, ensuring that their offerings accurately reflect current curriculum objectives and content. In doing so, they not only enable data-driven approaches to enhance and update their educational programs but also contribute to the overall effectiveness of AI-based curriculum development tools.

5.4. Building competitive and attractive curricula at peripheral universities

As demonstrated by Feresin (2023) and confirmed by the open data of the Italian Ministry of University and Research,[5] universities in southern or peripheral regions are losing thousands of students at an alarming rate. The demographic decline and the lesser attractiveness of Italian universities compared with other OECD countries (OECD/European Union, 2019) necessitate the development or new strategies. *How can peripheral universities be attractive and gain a competitive advantage?*

A recent study conducted in Italy by Columbu et al. (2021) reveals that the educational offerings (i.e. curricula) within and between universities – alongside socioeconomic and territorial conditions – can influence mobility. In some cases, attractiveness is more influenced by the university's vocation (defined as the number of curricula offered in a given field and their reputation in research quality within the same field of study) rather than the single curriculum. However, focusing on the educational offerings of peripheral universities is crucial only if they are not in direct competition with larger institutions. Findings from the literature clearly indicate that, overall, increasing diversification is beneficial to attract freshers, and in particular is vital for small or peripheral universities (Biagi et al., 2024). These types of HEIs should diversify their degree programs to enhance attractiveness, while deviating their portfolio from the national standard to maximize comparative advantage.

In rapidly evolving fields of study (like IEM), universities must continuously adapt their curricula and programs to remain current and

[5] https://ustat.mur.gov.it/opendata/.

narrow the skill gap between academia and industry. Curriculum developers, program chairs, and professors can use the approach proposed in this study as a reference for strategically constructing and positioning their educational offerings. The concept of AI as an ally in curriculum design would become particularly crucial for peripheral universities that may lack sufficient teaching support staff to conduct time-consuming analyses.

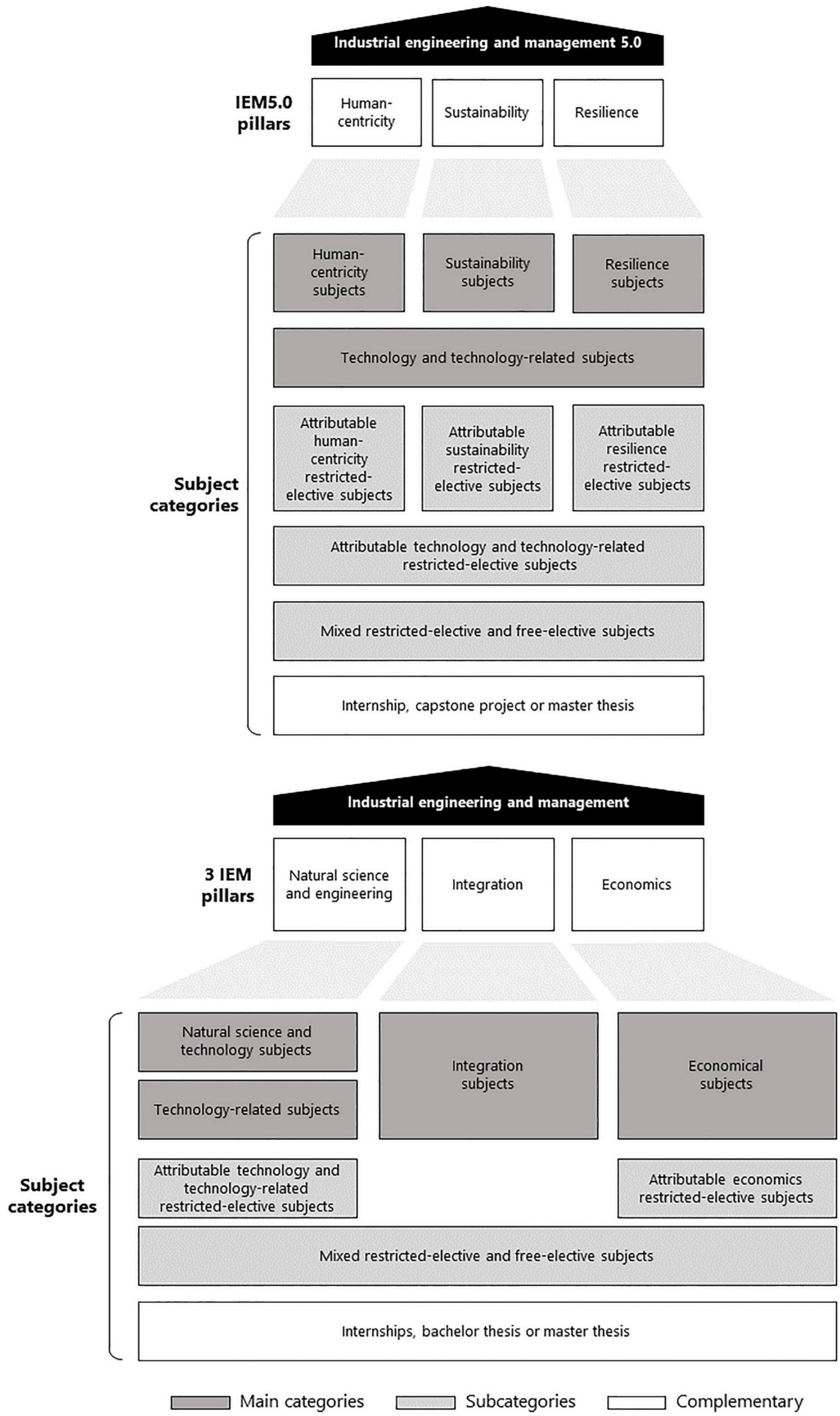

Fig. 6. Extending the WING model with IEM5.0 education.

5.5. Extending the IEM WING model towards industry 5.0

This study places a strong emphasis on advancing IEM education to provide a clear roadmap for curriculum development within the Industry 5.0 landscape. In our exploration, we have also underscored the critical aspect of positioning a curriculum within a specific discipline and identifying prerequisites to ensure students' successful journey. To further guide our curriculum development endeavors, we turned to the well-established WING model, recognized and endorsed by alumni associations in German-speaking countries (Zunk & Sadei, 2015). This model thoughtfully categorizes subjects into four distinct areas: natural science and technology subjects, technology-related subjects, economics, and integration subjects. Moreover, it provides a suggested balance between technical and economic subjects, advocating for an ideal ratio of 50–70% in favor of technical subjects. As illustrated in Fig. 6, one strategic move that emerged is the sequencing of an Industry 5.0 curriculum after an IEM curriculum. This approach offers notable advantages. Beginning with a robust foundation established through IEM studies, which encompass key areas like operations research, supply chain management, quality control, and project management, students gain a deep understanding of traditional industrial processes and management principles. This foundational knowledge serves as the basis for appreciating the relevance and applicability of Industry 5.0 technologies in real-world contexts. Moreover, the transition from IEM to Industry 5.0 cultivates interdisciplinary thinking, an essential skill for navigating the multifaceted challenges posed by Industry 5.0. In crafting a curriculum aligned with Industry 5.0, it is paramount to delineate subject categories informed by the WING model framework. These categories, informed by the WING model framework, serve as the educational pillars upon which the curriculum is constructed.

- **Technology Subjects**: These subjects form the bedrock of the curriculum, offering students essential knowledge in emerging technologies that underpin Industry 5.0. They encompass topics such as IoT implementation, artificial intelligence in manufacturing, cybersecurity for Industry 5.0, and automation.
- **Technology-Related Subjects**: Complementary to the core technology subjects, this category explores the broader implications of technology integration within Industry 5.0. It delves into topics like digital transformation strategies, technology adoption in supply chain management, and real-world case studies of Industry 5.0 implementations.
- **Human-Centricity Subjects**: Given the human-machine collaboration inherent to Industry 5.0, this category delves into the pivotal role of human factors and ethical considerations. Subjects within this category encompass human-machine interface design, ethical dilemmas in Industry 5.0, human factors in autonomous systems, and workplace ergonomics.
- **Sustainability Subjects**: Sustainability is paramount in the modern industrial landscape. This category focuses on environmentally responsible practices within Industry 5.0, covering topics such as sustainable manufacturing, green supply chain management, renewable energy integration, and eco-design principles.
- **Resilience Subjects**: Resilience is a cornerstone of Industry 5.0. This category addresses risk management, disaster recovery, and business continuity, offering subjects that explore business continuity planning, risk assessment in Industry 5.0, supply chain resilience, and crisis management.
- **Attributable Technology and Technology-Related Restricted-Elective Subjects**: Specialized technology or technology-related courses within specific domains offer students opportunities for deeper exploration. Examples include advanced data analytics for Industry 5.0, IoT applications in specific industries, and robotics and automation in specialized contexts.
- **Attributable Human-Centricity Restricted-Elective Subjects**: This category allows students to specialize in human-centric aspects of Industry 5.0, focusing on areas such as human-machine interaction, organizational culture transformation, and ethics within advanced technological contexts.
- **Attributable Sustainability Restricted-Elective Subjects**: Students can delve further into sustainability within Industry 5.0, with topics such as sustainable materials in manufacturing, carbon footprint reduction strategies, and circular economy principles.
- **Attributable Resilience Restricted-Elective Subjects**: Specialized resilience subjects provide advanced knowledge in areas like cybersecurity resilience, supply chain risk modeling, and crisis communication strategies.
- **Mixed Restricted-Elective and Free-Elective Subjects**: These subjects offer a blend of structured restricted electives and flexible electives, enabling students to tailor their curriculum while maintaining core competencies. Examples include mixed supply chain management electives, interdisciplinary project courses, and technology innovation electives.
- **Internship, Capstone Project, or Master Thesis**: These components bridge theory and practice, providing students with opportunities to apply knowledge and conduct research in collaboration with industry partners. Examples encompass Industry 5.0 internships, capstone projects focusing on smart manufacturing, and master's theses on sustainability practices in Industry 5.0.

The seamless alignment of these educational pillars and the strategic sequencing of IEM and Industry 5.0 curricula lay the foundation for comprehensive, forward-thinking, and highly relevant engineering education.

6. Conclusions

In this paper, we present a critical need for education of systems-level thinkers in the global manufacturing industry, evidenced by the relatively recent digital transformation of enterprises. We showcase a novel AI-powered curriculum development system which helps educators to build cutting-edge learning paths and speed up/automate the most time-intensive phases of the curriculum building and topic selection process. In brief, the system aids curriculum developers to: 1) identify the relevant and cutting-edge education areas and learning topics; 2) evaluate similarities and differences with other education institutes to position the educational offer coherently in the (inter-)national panorama; 3) pick and map learning topics within an innovative curriculum. We believe that such a system not only helps curriculum developers to define and maintain their educational offer, but also empowers learners to being up-to-date on desirable knowledge and help them to clarify the vision of the future job they want to opt for.

Based on the findings of this study, we can identify promising research directions. First, we encourage further exploration of AI applications in curriculum design, for example to find prerequisites within a curriculum and establish learning pathways (Vuong et al., 2011). This could involve developing AI algorithms to analyze course content and structure to optimize curriculum coherence and synergy. This would also expand the use of AI to develop 'curriculum maps' as tools to evaluate the consistency of a degree as a whole (Icarte & Labate, 2016). Second, we argue the potential of AI to assist educators in improving teaching practices, developing tailored instructional materials, and providing real-time feedback on performance. This could involve designing AI-driven professional development programs and studying their effectiveness in enhancing teaching quality and student learning outcomes. Third, we believe it is worthy investigating the effects of programmatic diversity on HEIs' attractiveness, including factors driving conversion of non-students to students, student retention until graduation, and flows between universities. In line with these directions, future work will be structured in two streams: on one hand, we will assess the effectiveness of the 'digital twin engineering curriculum based on questionnaires that will be submitted to the students at the end of the

Professional Master Degree Program in Industry of the Future conducted at the University of Calabria; on the other hand, we will design and develop a curriculum for human-centric industrial engineering and management as part of the project LEONARDO (Learning & Experimentation OpeN-Access factoRy for inDustrial wOrkforce 5.0) funded under the key action 2 of the call 2023 – Erasmus+ programme. We expect this future work on integrating AI-driven innovations into curriculum design and teaching practices to optimize curriculum coherence and create dynamic learning pathways, thus enhancing the educational experience and outcomes for students and increasing the competitiveness and appeal of our educational offerings.

List of acronyms

Acronym	Definition
AI	Artificial Intelligence
BERT	Bidirectional Encoder Representations from Transformers
CATAR	Content Analysis Toolkit for Academic Research
CBC	Competency-based curriculum
CBE	Competency-based education
GPT	Generative Pre-trained Transformer
HEI	Higher education istitution
HTT	Hierarchical topic tree
IEM	Industrial Engineering and Management
IT/OT	Information Technology/Operational Technology
IoT	Internet of Things
ML	Machine Learning
NLP	Natural Language Processing
OECD	Organisation for Economic Co-operation and Development
PCA	Principal Component Analysis
SATI	Statistical Analysis Toolkit for Informetrics
SBERT	Sentence Bidirectional Encoder Representations from Transformers
SEP	Scientific evolutionary pathways
SUA-CDS	Scheda Unica Annuale(Italian)
TF-IDF	Term Frequency–Inverse Document Frequency
t-SNE	t-distributed Stochastic Neighbor Embedding

(*continued on next column*)

(*continued*)

CRediT authorship contribution statement

Antonio Padovano: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Martina Cardamone:** Writing – original draft, Visualization, Software, Resources, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A

Table A.1
ERASMUS + projects considered in this study

Project Code	Project title
2020-1-BG01-KA203-079025	Adaptation of strategies for corporate social responsibility to address the implications of the Industry 4.0
2019-1-MT01-KA203-051265	An Innovative Higher Education Institution Training Toolbox to EffeCtively AddRess the EUropean InduStry 4.0 Skills Gap and Mismatches
2019-1-PL01-KA203-065731	Building next generation competencies for logisticians and supply chain managers
2017-1-SI01-KA203-035522	Capitals of Smart Product Development
2019-1-RO01-KA203-063153	Development of mechatronics skills and innovative learning methods for Industry 4.0
2019-1-RO01-KA203-063486	Digital Manufacturing Master Degree to set specialists for the dawn of the Industry 4.0
2020-1-ES01-KA226-HE-095291	Digital Training for Cybersecurity Students in Industrial Fields
2019-1-ES01-KA203-065796	INCOBOTICS - Ready for Industry 5.0
2018-1-ES01-KA203-050493	Industrial Cyber Security 4.0
2019-1-SE01-KA203-060572	Manufacturing Education for a Sustainable fourth Industrial Revolution
2017-1-PL01-KA203-038698	Modern logistics learning: Certified module on master study level
2016-1-ES01-KA203-024921	Open Source Applications for Industrial Automation
2018-1-EE01-KA203-047094	Virtual Learning Factory Toolkit
2018-1-TR01-KA203-059739	Curriculum Development for Rapid Prototyping in Engineering Education
2019-1-NL01-KA203-060501	Purchasing Education Research Syndicate: Industry4.0 Skills Transfer
2017-1-SE01-KA203-034524	Social Network based doctoral Education on Industry 4.0
2017-1-IT02-KA203-036980	SPRINT4.0 - Strategic PaRtnership for INdustry 4.0 innovation advanced Training
2017-1-HR01-KA203-035408	The ICT Engineer of the 21st Century: Mastering Technical Competencies, Management Skills, and Societal Responsibilities

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